

Marko Cvjetko

PhD Student

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I am a 2nd year PhD student supervised by [Pierre-Yves Oudeyer](#) and [Clément Moulin-Frier](#), developing curiosity-driven AI for scientific discovery. I draw on goal-conditioned reinforcement learning, intrinsically motivated exploration, vision–language models, and evolutionary search to build agents that autonomously explore, characterize, and control complex systems. I apply these methods to artificial life substrates — such as cellular automata — to study open-ended evolution and self-organizing phenomena. I am extending this program toward deeper integration of LLMs as discovery partners in the experimental loop, and toward more biologically grounded substrates through an ongoing collaboration with the [Levin Lab](#). I am a 2026 [DISI](#) fellow and will visit [Clément Hongler's group](#) at EPFL in May 2026.

PUBLICATIONS

Under review

The Artificial Experimentalist: Discovery and Control of Self-Organizing Phenomena with Autotelic Reinforcement Learning. M. Cvjetko, B. Hartl, M. Levin, C. Moulin-Frier, P.-Y. Oudeyer. Submitted to *ALIFE 2026*. [\[project page\]](#). [\[code\]](#). *Goal-conditioned RL agent that discovers and controls self-organizing patterns in cellular automata via minimal local interventions, with zero-shot generalization to unseen dynamics and action spaces.*

Journal (invited, in preparation)

Extension of an *ALIFE 2025* paper for the *Artificial Life* journal special issue.

Conference papers

Exploring Flow-Lenia Universes with a Curiosity-driven AI Scientist: Discovering Diverse Ecosystem Dynamics. T. Michel, M. Cvjetko (co-first authors), G. Hamon, P.-Y. Oudeyer, C. Moulin-Frier. In *Proc. ALIFE 2025*, pp. 633–643. [doi:10.1162/ISAL.a.896](#). [\[project page\]](#). *Population-based diversity search (IMGEPs) over system-wide complexity and evolutionary-activity metrics, surfacing ecosystem-level dynamics in Flow-Lenia.*

Expedition & Expansion: Leveraging Semantic Representations for Goal-Directed Exploration in Continuous Cellular Automata. S. Khajehabdollahi, G. Hamon, M. Cvjetko, P.-Y. Oudeyer, C. Moulin-Frier, C. Colas. In *Proc. ALIFE 2025*. [doi:10.1162/ISAL](#). [\[project page\]](#). *Hybrid exploration alternating novelty search with VLM-generated linguistic goals, providing evidence that LLM-generated goals act as stepping stones into unexplored regions of the behavioral space.*

Workshop paper

Evolving Symbiosis, from Barricelli's Legacy to Collective Intelligence: a Simulated and Conceptual Approach. J. Ashford, M. Cvjetko, R. Löffler, B. Sakallıoglu, A. Valerio, M. Tataryn, B. Hartl, L. Pio-Lopez, S. Nichele. Accepted to *GECCO 2026 Workshop Proceedings*. [arXiv:2603.08463](#). Runner-up award at [ALICE 2026 Workshop](#), Copenhagen. *Revisits Barricelli's symbiogenesis simulations as a substrate for studying the emergence of cooperation and collective intelligence.*

Late-breaking abstract

Discovering and Controlling Diverse Self-Organised Patterns in Cellular Automata Using Autotelic Reinforcement Learning. M. Cvjetko, G. Hamon, C. Moulin-Frier, P.-Y. Oudeyer. *ALIFE 2025*, Kyoto, Japan. [\[HAL\]](#). [\[project page\]](#)

HONORS, FELLOWSHIPS & RESEARCH VISITS

Diverse Intelligences Summer Institute (DISI) Fellow

June 2026

- Selected for a 3-week transdisciplinary residency on the origins, nature, and future of intelligences (cognitive science, AI, biology, philosophy); cohort of ~40 fellows funded by the John Templeton Foundation.

Funded research visit, group of Clément Hongler (EPFL)

May 2026

- Two-week visit funded by the host group, coinciding with the [Demeco 2026](#) workshop.

ALICE Workshop Runner-up Award, Copenhagen

Feb 2026

- Awarded for “Evolving Symbiosis, from Barricelli's Legacy to Collective Intelligence”.

EDUCATION

University of Bordeaux, Inria, Flowers Team, France

PhD thesis – Studying the emergence of open-ended evolution in cellular automata using curiosity-driven AI

Oct 2024 –
current

- **Supervised by** [Pierre-Yves Oudeyer](#) and [Clément Moulin-Frier](#)
- Designing AI tools for assisted scientific discovery: agents that autonomously discover, characterize, and steer self-organizing phenomena, applied to Artificial Life substrates such as [Lenia](#) and the study of open-ended evolution.
- **Supervision:** co-supervising a Master's intern on the emergence of memory and learning in cellular automata, and two BSc students on a project investigating Turing-completeness in [Lenia](#).
- **Service:** reviewer for *ALIFE 2026*.
- **Collaborations:** ongoing collaboration with Levin Lab (Tufts University) on autotelic agents for the discovery and control of self-organizing patterns; upcoming 2-week funded research visit hosted by [Clément Hongler \(EPFL\)](#), coinciding with the [Demeco 2026](#) workshop (May 2026).

University of Zagreb, Faculty of Electrical Engineering and Computing (FER), Croatia

MSc, Computer Science

Sep 2021 – Jul
2024

- **MSc thesis:** Recognition of the respiratory signal from thermal video.
Supervised by [Prof. Sebastian Haesler](#) and [Prof. Siniša Šegvić](#)
In collaboration with [Neuro-Electronics Research Flanders \(imec/KU Leuven/VIB\)](#), project described in [work experience](#) section.
- **Selected courses:** *Machine Learning, Deep Learning, Soft Computing, Parallel Programming*

Erasmus student exchange at KU Leuven, Flanders, Belgium

Sep 2022 – Jul
2023

- Coursework from the [Advanced Master AI](#) programme
- **Selected courses:** *Cognitive Science, Philosophy of Mind, Linguistics and AI Uncertainty in AI, Neural Computing,*

BSc, Computing

Sep 2016 – Sep
2021

- **BSc thesis:** *Music Synthesis using Artificial Intelligence* – LSTM next-note prediction on symbolically represented music. Supervised by [Prof. Domagoj Jakobović](#).

WORK EXPERIENCE

Graduate Researcher, [Neuro-Electronics Research Flanders \(imec/KU Leuven/VIB\)](#), [Haesler Lab](#) ([GitHub](#))

Mar 2023 – Jul
2024

- **Supervised by** [Prof. Sebastian Haesler](#)
- Developed a machine learning framework for extracting and analyzing respiratory responses of head-restrained mice from infrared footage, in collaboration with experimental neuroscientists at Haesler Lab.
- Tackled low-data and domain-shift challenges via unsupervised and self-supervised learning with vision transformers and deep CNNs.
- Configured the data annotation platform, oversaw annotation tasks and trained students on annotation guidelines. Presented in a [poster session](#) at the institute's yearly retreat.

Machine Learning Engineer Internship, [DoXray](#)

Jun 2022 – Sep
2022

- Fine-tuned LayoutXLM — a multilingual multimodal (text + layout + vision) transformer — for named-entity recognition on German invoice documents, benchmarked against a BERT baseline. [Blogpost](#).

PROJECTS

Generative models for unsupervised behavioral spaces in open-ended discovery	2026 – current
• Training diffusion models and variational autoencoders to represent complex system behavior and sample their parameters for open-ended diversity search • Goal: replace hand-designed semantic descriptors with learned, generative behavior spaces that enable scalable goal-directed search and quality–diversity over continuous cellular automata.	
Music-reactive cellular automata	Nov 2024
• Real-time music-driven Lenia animations: tempo and frequency features drive a continuous cellular automaton (example). Built during the hack1robo hackathon.	
Cellular Automata Study in Jax (GitHub)	Jan – Mar 2024
• Lenia implementation in Jax; trained a 13-parameter neural network to learn Conway's Game of Life update rule.	
Automatic Universal Taxonomies for Multi-Domain Semantic Segmentation (GitHub)	2023 – 2024
• Joint label space for semantic segmentation datasets via cross-dataset confusion matrices, enabling to train a single classification head on the union.	
Can Augmentation Make Better Doctors? Data Augmentation in a Low-Resource Sequence Labeling Task	2022
Benchmarked data augmentations for BERT-based sequence labeling on a low-resource medical named entity recognition dataset.	

SKILLS

Programming	Python, C, Bash
ML / DL	PyTorch, JAX, NumPy, Hugging Face Transformers, Optuna
Reinforcement learning	Stable-Baselines3, CleanRL
Evolutionary / QD search	QDax
LLM / VLM serving	vLLM
HPC	Slurm, Singularity (parallel multi-node, multi-GPU workers, asynchronous coordination)
Tooling	Linux, Docker, Git, LaTeX
Languages	Croatian (native), English (fluent), French (intermediate)